

**ITCC103 (IT Fundamentals)
Week 5 Assignment**

Module 2 – Unit 1 & Unit 2 (Using Data Types and Units & Using Apps)

Name: No Group

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Subject Code-Section: ITCC103-102I

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1. Explain the relationship between binary notation and the physical components of a computer, such as transistors. Why is binary notation the fundamental language of computers, and how does this concept apply to the digital representation of all types of data, including text, images, and sound?

Answer: Transistors act like tiny switches that are either on or off. Binary uses the same idea: 1 means on, 0 means off. This is why computers use binary, it matches how the hardware works. All data, like text, images, and sound, is turned into long strings of 0s and 1s so the computer can store and process it.

2. Our lesson describes hexadecimal notation as a more efficient way to represent large values than binary. Describe a real-world scenario in which hexadecimal notation is used, and explain why it is more practical for humans to use in that context.

Answer: Hexadecimal is used in color codes for websites, like #FF5733. Writing the same value in binary would be very long and hard to read. Hex is shorter and easier for people to work with, which is why it's used for colors, memory addresses, and similar things.

3. Our lesson discusses several forms of intellectual property, including copyright, trademark, and patent. Choose one of these forms of protection and explain a time when you have encountered it in your daily life, such as in a software application, a logo, or a product design.

Answer: I see the Nike "swoosh" logo often on shoes and clothes. It is trademarked, so no other brand can copy it. This protects Nike's identity and stops people from getting confused about who made the product.

4. Our lesson states that data is used to drive strategic business decisions in areas like sales and marketing. Provide a specific example of how a company could use data analytics to improve its marketing campaign for a product you are familiar with.

Answer: A coffee shop could look at app data and notice that customers buy more coffee in the afternoon. They could then send a discount notification in the afternoon to get more sales, and check later if it worked.

5. Explain the difference between a bit, a byte, a kilobyte, and a gigabyte, and provide a real-world example of a file that would be measured in each of these units. Why is it important to understand these units when dealing with a computer's storage capacity?

Answer: A bit is just one 0 or 1. A byte is 8 bits, enough for one letter. A kilobyte is about 1,000

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bytes, like a small text file. A gigabyte is about 1 billion bytes, like a movie file. Knowing these units helps you understand how much space a file needs and if it will fit on your device.

6. Our lesson describes various types of software licenses, including proprietary, freeware, shareware, and open source. Choose two of these license types and explain how a developer's choice of license could impact both their business model and a user's experience with the software.

Answer: Proprietary software, like Microsoft Office, is closed and paid. The company earns money from subscriptions, and users get support but must pay. Open source software, like LibreOffice, is free and anyone can see or change the code. Developers usually earn money through support services instead, and users get free access but less guaranteed support.

7. Explain the importance of a proper uninstallation process for an application. What are the potential consequences of simply deleting an application's files from the Program files folder and how does this relate to the concept of "orphaned" files and registry entries?

Answer: Uninstalling properly removes not just the program but also its registry entries and extra files. If you just delete the folder, those leftover files and entries stay behind this is called "orphaned" data. This can slow down the computer, waste space, and cause problems if you try to reinstall the program later.

8. Our lesson discusses several best practices for installing software, such as checking for OS and hardware compatibility. Describe a hypothetical situation where a user ignores these practices, and explain the specific problems they might encounter as a result.

Answer: If someone installs a program that needs 8GB RAM on a computer with only 4GB, the program might not install at all, or it could crash a lot, run very slowly, or cause errors because the computer can't handle it properly.

9. Our lesson categorizes software into different types, such as productivity, collaboration, and business software. Choose one specific application (e.g., Google Docs, Microsoft Teams) and describe how it can be used for both productivity and collaboration, providing examples of its features in each context.

Answer: Google Docs can be used alone for productivity, like writing a report using spell-check and templates. It can also be used for collaboration, where many people edit the same document together, leave comments, and track changes in real time.

10. What is the purpose of an End-User License Agreement (EULA)? Describe a scenario where a user might encounter a EULA, and explain why it is a legally binding contract that defines the terms of use for the software.

Answer: A EULA is a rule agreement between the software company and the user about how the software can be used. A common example is when installing a program and being asked to click "I Agree" before continuing. It is legally binding because agreeing to it means you accept the rules, and breaking them can lead to consequences like losing access to the software.